

Digital Forensics Laboratory

EXPERIMENT NO: 01

DATE: August 9, 2026

TITLE: Comprehensive Digital Evidence Acquisition and Command-Line Analysis
using FTK Imager, WinPmem, and Volatility

1. AIM

To perform a complete forensic acquisition and analysis lifecycle by imaging non-volatile storage using AccessData FTK Imager, capturing live volatile memory (RAM) using the WinPmem command-line utility, and analyzing the extracted memory dump using Volatility 3 commands via the PassMark Volatility Workbench.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows OS
- **Forensics Tools:**
 - AccessData FTK Imager (GUI for non-volatile physical drive imaging)
 - WinPmem (winpmem_mini_x64_rc2.exe) (CLI tool for volatile memory capture)
 - Volatility 3 Framework (vol.exe / PassMark Volatility Workbench V3.0)
- **Hardware:** Evidence PC/Workstation, Target Storage Drive (hp v221w 16GB USB Device).

3. THEORY

A thorough digital forensic investigation requires collecting both data at rest and data in motion using strict operational procedures.

- **Non-Volatile Memory Acquisition:** AccessData FTK Imager creates mathematically verifiable, bit-by-bit physical images of storage devices (saved as Raw/dd) without altering the original evidence.

- **Volatile Memory (RAM) Acquisition:** Command-line utilities like **WinPmem** load a specialized kernel driver (pme98A2.tmp) to read physical memory directly. Executing this via CMD writes active processes, network connections, and decrypted keys to a raw dump file (.raw).
- **Volatile Memory Analysis (CLI): Volatility 3** is a framework for analyzing RAM dumps. By passing specific parameters and plugins (e.g., -f <image_path> windows.pslist.PsList) into the command line or workbench, investigators can reconstruct the system's live state and identify suspicious processes.

4. PROCEDURE

Part A: Acquiring Non-Volatile Memory (Disk Image) using FTK Imager Step 1: Select the Source Type

Open FTK Imager as an Administrator. Navigate to **File > Create Disk Image**. In the Select Source dialog, choose **Physical Drive**.

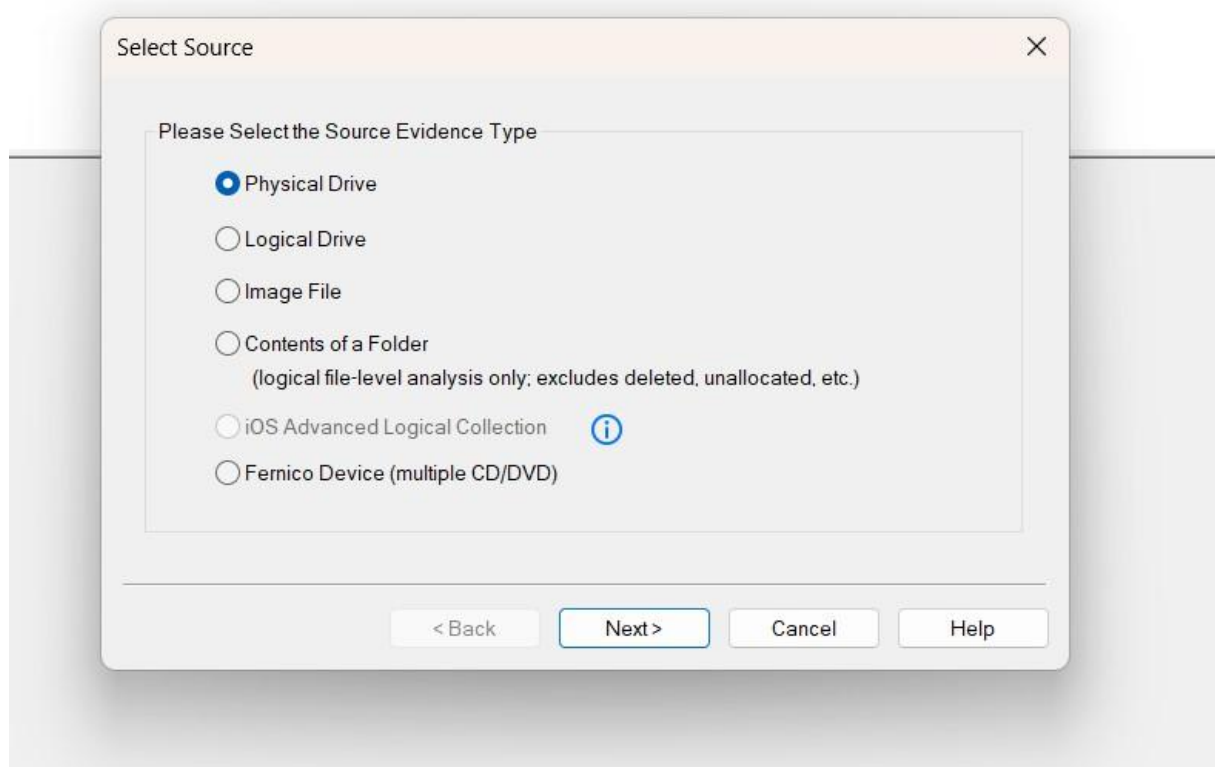


Figure 1: Selecting Physical Drive as the source evidence type

Step 2: Select the Specific Drive

Select the target evidence drive (\\.\PHYSICALDRIVE1 - hp v221w USB Device [16GB USB]) from the dropdown.

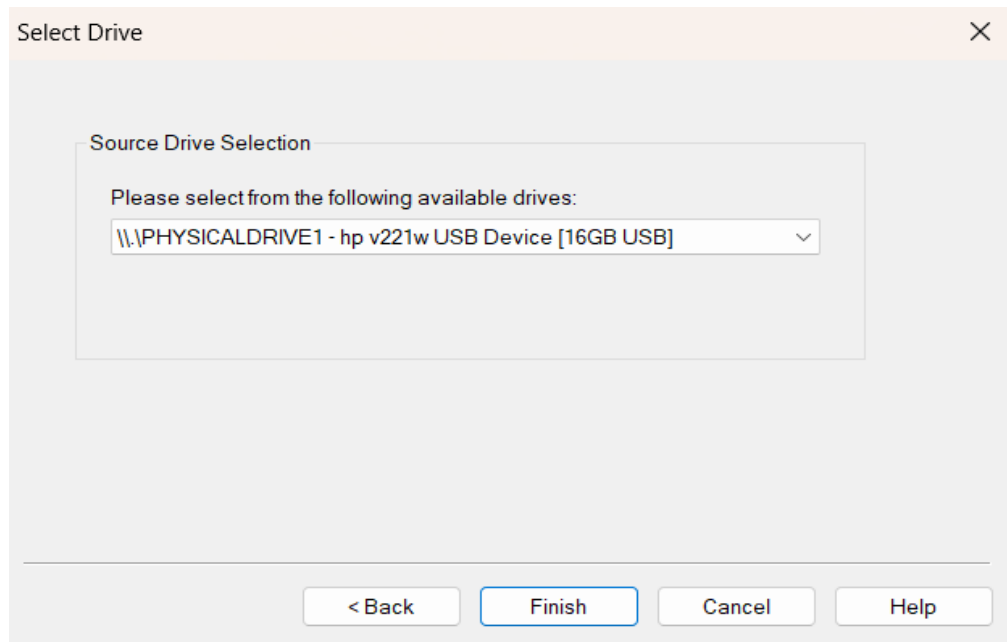


Figure 2: Selecting the target USB device for acquisition

Step 3: Choose the Image Type

Select **Raw (dd)** as the destination image type to create a pure bitstream copy.

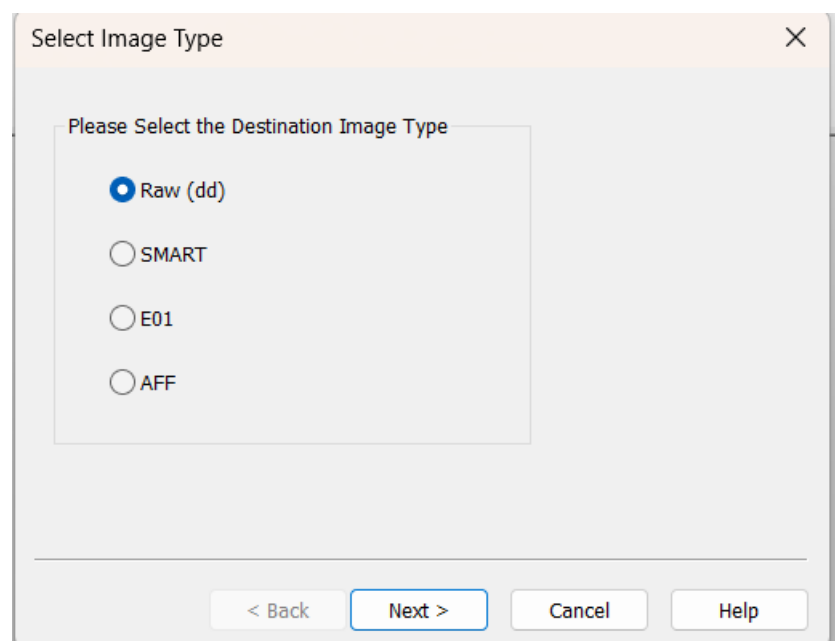
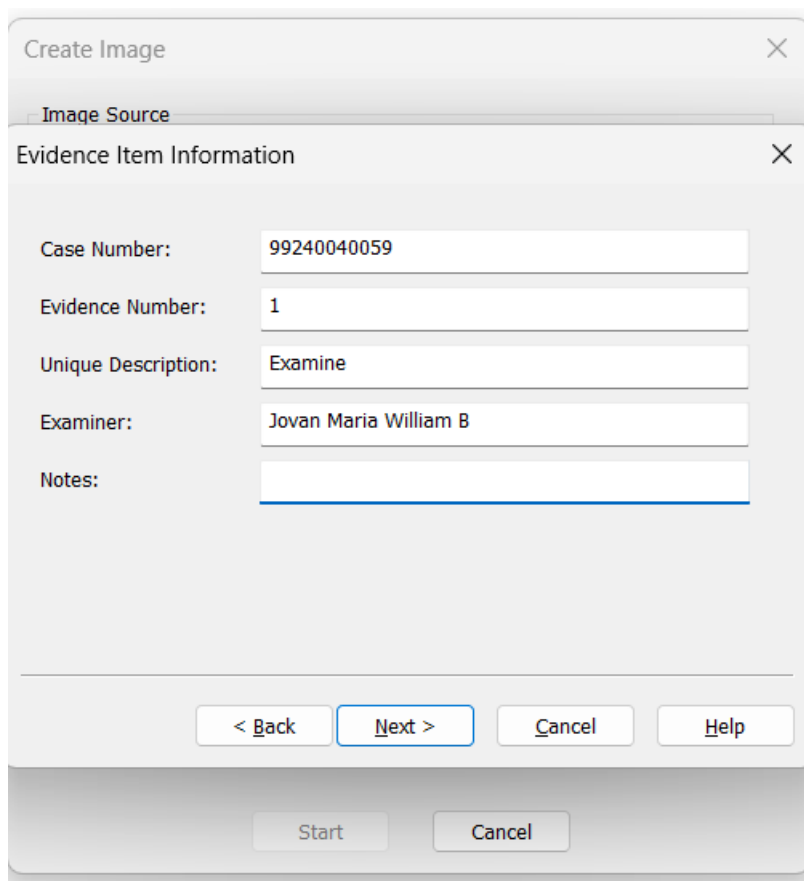


Figure 3: Configuring the destination image format to Raw (dd)

Step 4: Enter Evidence Item Information

Provide the standardized case details to maintain the chain of custody:

- **Case Number:** 99240040059
- **Evidence Number:** 1
- **Unique Description:** Examine
- **Examiner:** JovanMariaWilliam B



The screenshot shows a 'Create Image' dialog box with a tab labeled 'Evidence Item Information'. The dialog contains several text input fields for case details. The 'Case Number' field is filled with '99240040059', 'Evidence Number' with '1', 'Unique Description' with 'Examine', and 'Examiner' with 'Jovan Maria William B'. The 'Notes' field is empty. At the bottom of the dialog, there are four buttons: '< Back', 'Next >', 'Cancel', and 'Help'. The 'Next >' button is highlighted with a blue border. Below the dialog box, there are two more buttons: 'Start' and 'Cancel'.

Case Number:	99240040059
Evidence Number:	1
Unique Description:	Examine
Examiner:	Jovan Maria William B
Notes:	

< Back Next > Cancel Help

Start Cancel

Step 5: Configure Image Destination

Set the destination folder to A:\DF\Digital_Forensics_Lab\Experiment-01-FTK-Imager\Results and the filename to Result. Set the Fragment Size to 20 MB and check **Verify images after they are created**.

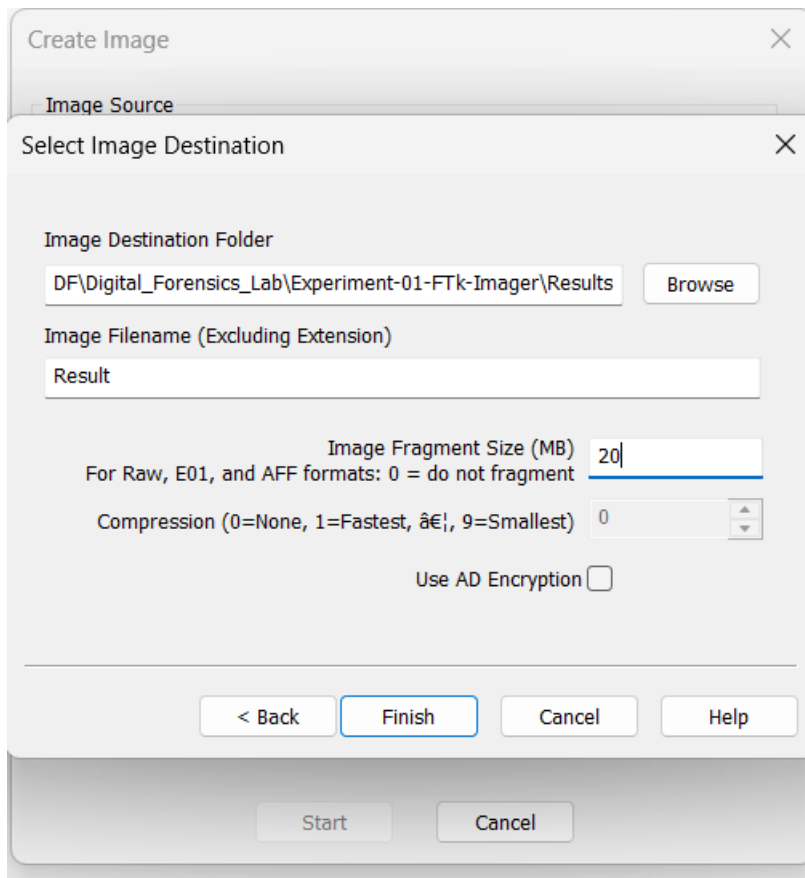


Figure 5: Setting the destination path, filename, and fragmentation size

Step 6: Monitor Acquisition Progress & Hash Verification

Click **Start**. Once complete, FTK Imager hashes the original drive and the newly created image. Review the Drive/Image Verify Results.

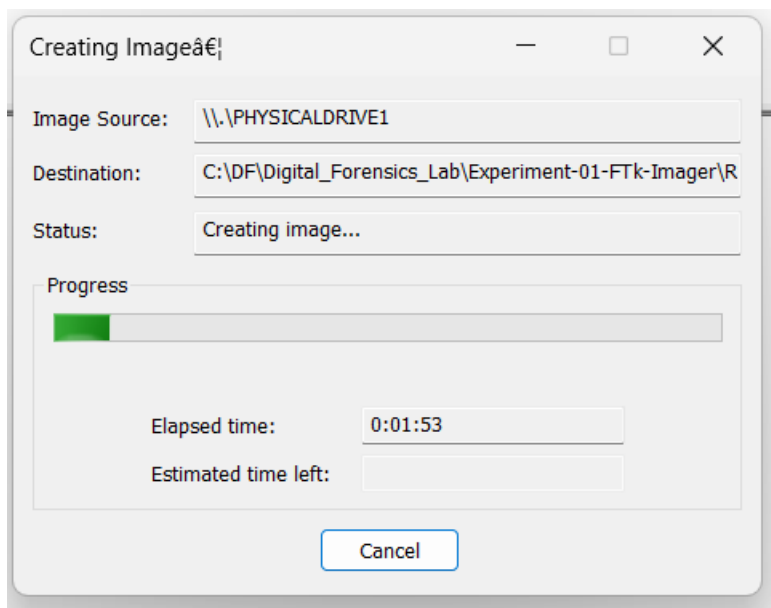


Figure 6: Disk imaging progress tracking

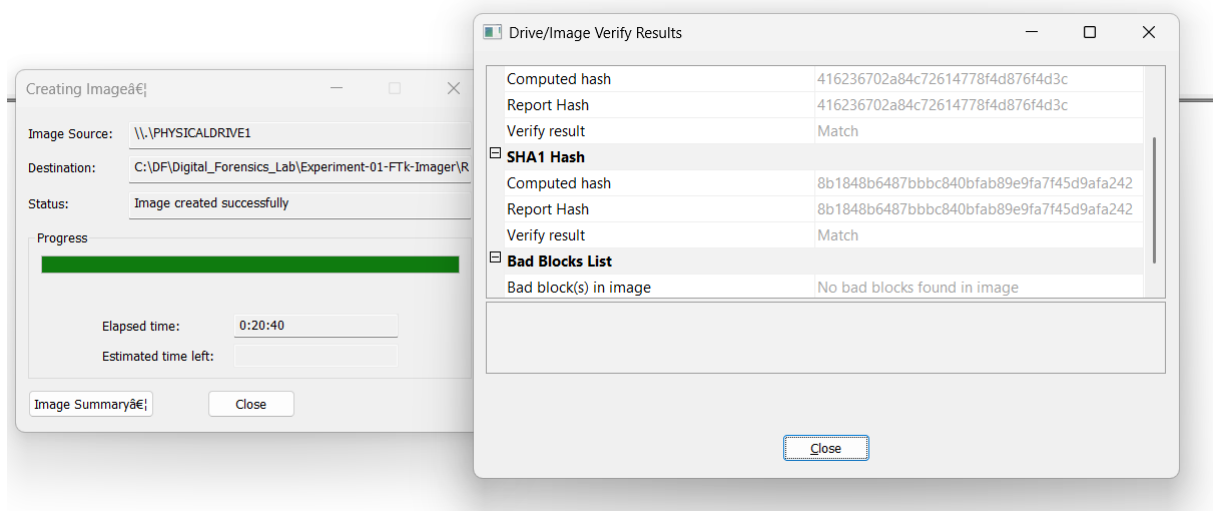


Figure 7: Successful verification of MD5 and SHA1 hashes

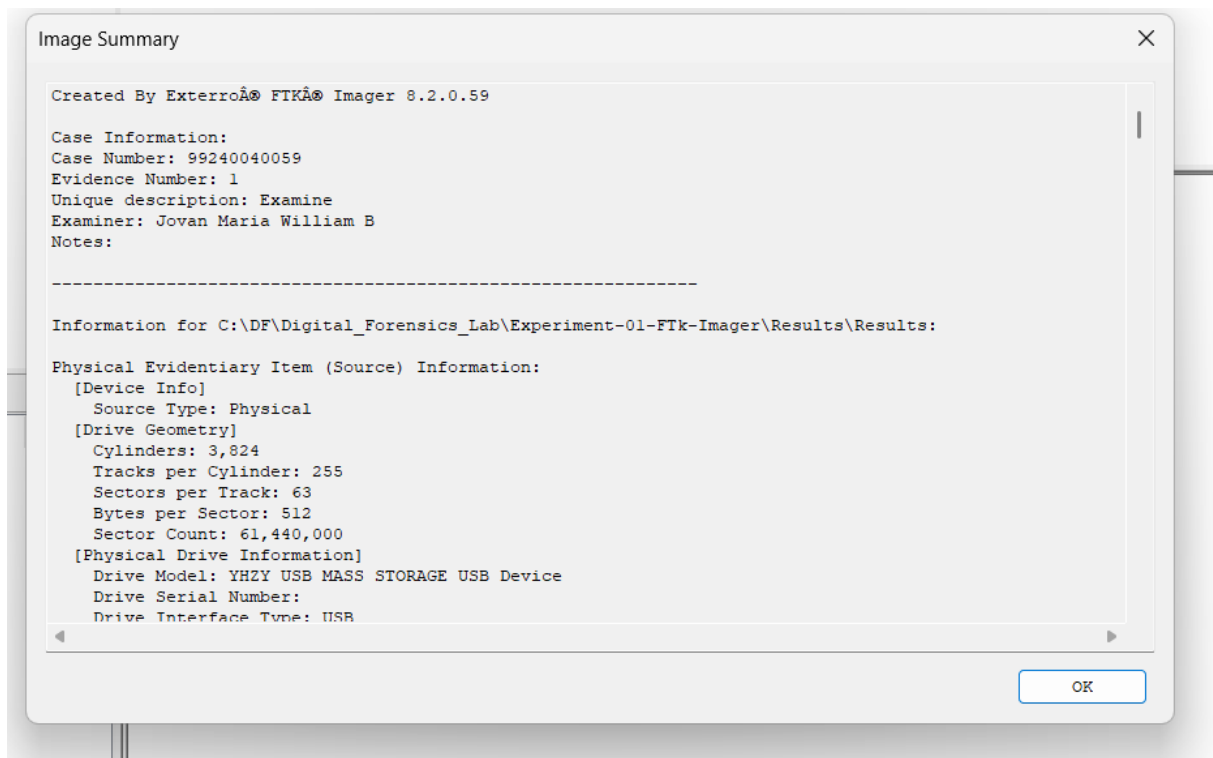


Figure 8: Image Summary detailing the physical device geometry

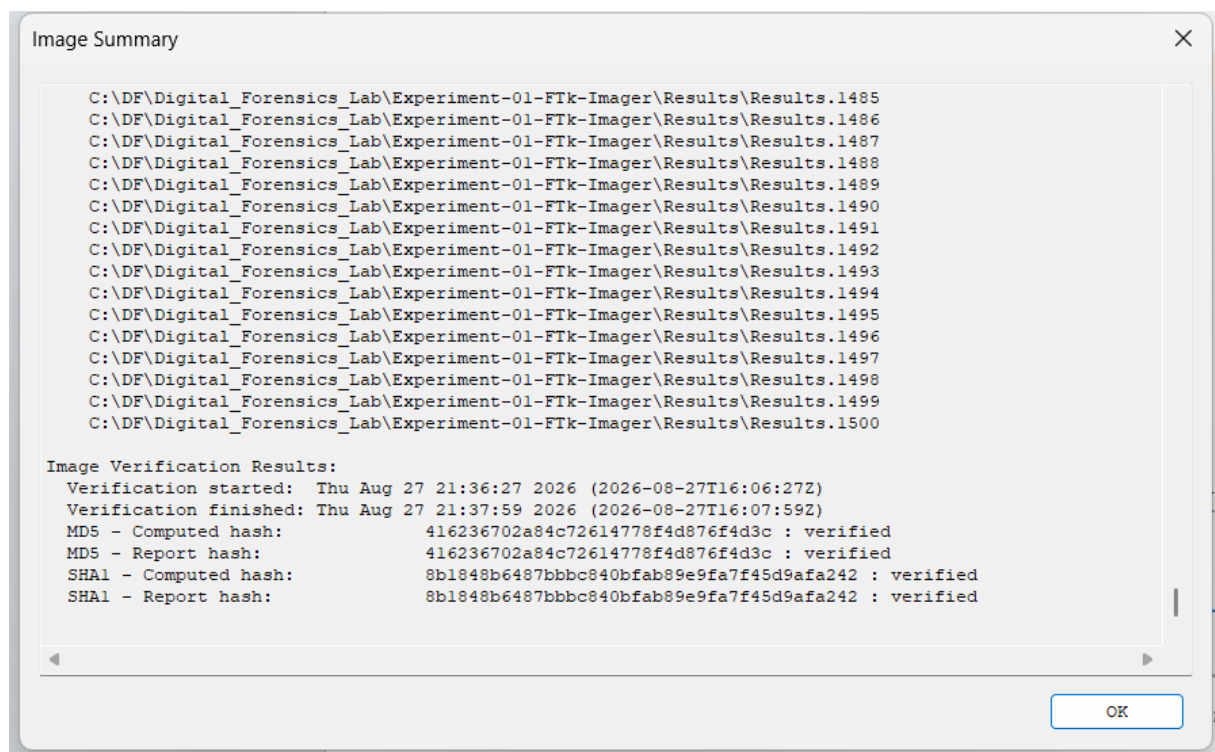


Figure 9: Image Summary detailing the hash verification results and segment list

Part B: Acquiring Volatile Memory (RAM) using WinPmem CMD Step 1: Launch Command Prompt

Press Win + R, type cmd, and press Ctrl + Shift + Enter to open the Command Prompt as an Administrator.

Step 2: Navigate to Directory

Use the cd command to navigate to the folder containing the WinPmem executable:

DOS

C:

Cd\DF

Step 3: Check Available Parameters

Execute the help command to view the WinPmem usage parameters: DOS

winpmem_mini_x64_rc2.exe -h

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd \DF
C:\DF>cd Digital_Forensics_Lab
C:\DF\Digital_Forensics_Lab>cd Experiment-01-Ftk-Imager
C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager>dir
Volume in drive C is Windows
Volume Serial Number is 0A2E-2A00

Directory of C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager

27-08-2026 09:14 PM <DIR>      .
27-08-2026 09:14 PM <DIR>      ..
27-08-2026 10:06 PM <DIR>      Results
0 File(s)            0 bytes
3 Dir(s) 90,353,725,440 bytes free

C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager>cd Results
C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager\Results>dir
Volume in drive C is Windows
Volume Serial Number is 0A2E-2A00

Directory of C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager\Results

27-08-2026 10:06 PM <DIR>      .
27-08-2026 09:14 PM <DIR>      ..
27-08-2026 09:15 PM      20,971,520 Results.001
27-08-2026 09:37 PM      119,491 Results.001.txt
27-08-2026 09:15 PM      20,971,520 Results.002
27-08-2026 09:15 PM      20,971,520 Results.003
27-08-2026 09:15 PM      20,971,520 Results.004
27-08-2026 09:15 PM      20,971,520 Results.005
27-08-2026 09:15 PM      20,971,520 Results.006
27-08-2026 09:15 PM      20,971,520 Results.007
27-08-2026 09:15 PM      20,971,520 Results.008
27-08-2026 09:15 PM      20,971,520 Results.009
27-08-2026 09:15 PM      20,971,520 Results.010
27-08-2026 09:15 PM      20,971,520 Results.011
27-08-2026 09:15 PM      20,971,520 Results.012
27-08-2026 09:15 PM      20,971,520 Results.013
27-08-2026 09:15 PM      20,971,520 Results.014
27-08-2026 09:15 PM      20,971,520 Results.015
27-08-2026 09:15 PM      20,971,520 Results.016
27-08-2026 09:15 PM      20,971,520 Results.017
```

Step 4: Execute Memory Capture Command

Run the memory extraction command, specifying the output filename Inba.raw. The utility will map the physical memory and write it to the file:

DOS

winpmem_mini_x64_rc2.exe

Step 5: Verification of Dump

Wait for the terminal to display Driver Unloaded. Navigate to the directory to verify the 18.2

Administrator: C:\Windows\system32\cmd.exe

```
27-08-2026 09:29 PM 20,971,520 Results.961
27-08-2026 09:29 PM 20,971,520 Results.962
27-08-2026 09:29 PM 20,971,520 Results.963
27-08-2026 09:29 PM 20,971,520 Results.964
27-08-2026 09:29 PM 20,971,520 Results.965
27-08-2026 09:29 PM 20,971,520 Results.966
27-08-2026 09:29 PM 20,971,520 Results.967
27-08-2026 09:29 PM 20,971,520 Results.968
27-08-2026 09:29 PM 20,971,520 Results.969
27-08-2026 09:29 PM 20,971,520 Results.970
27-08-2026 09:29 PM 20,971,520 Results.971
27-08-2026 09:29 PM 20,971,520 Results.972
27-08-2026 09:29 PM 20,971,520 Results.973
27-08-2026 09:29 PM 20,971,520 Results.974
27-08-2026 09:29 PM 20,971,520 Results.975
27-08-2026 09:29 PM 20,971,520 Results.976
27-08-2026 09:29 PM 20,971,520 Results.977
27-08-2026 09:29 PM 20,971,520 Results.978
27-08-2026 09:29 PM 20,971,520 Results.979
27-08-2026 09:29 PM 20,971,520 Results.980
27-08-2026 09:29 PM 20,971,520 Results.981
27-08-2026 09:29 PM 20,971,520 Results.982
27-08-2026 09:29 PM 20,971,520 Results.983
27-08-2026 09:29 PM 20,971,520 Results.984
27-08-2026 09:29 PM 20,971,520 Results.985
27-08-2026 09:29 PM 20,971,520 Results.986
27-08-2026 09:29 PM 20,971,520 Results.987
27-08-2026 09:29 PM 20,971,520 Results.988
27-08-2026 09:29 PM 20,971,520 Results.989
27-08-2026 09:29 PM 20,971,520 Results.990
27-08-2026 09:29 PM 20,971,520 Results.991
27-08-2026 09:29 PM 20,971,520 Results.992
27-08-2026 09:29 PM 20,971,520 Results.993
27-08-2026 09:29 PM 20,971,520 Results.994
27-08-2026 09:29 PM 20,971,520 Results.995
27-08-2026 09:29 PM 20,971,520 Results.996
27-08-2026 09:29 PM 20,971,520 Results.997
27-08-2026 09:29 PM 20,971,520 Results.998
27-08-2026 09:29 PM 20,971,520 Results.999
27-08-2026 10:05 PM 527,640 winpmem_mini_x64_rc2.exe
1502 File(s) 31,457,927,131 bytes
2 Dir(s) 90,353,225,728 bytes free
```

C:\DF\Digital_Forensics_Lab\Experiment-01-FTk-Imager\Results>

```

Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>cd \DF

C:\DF>cd Digital_Forensics_Lab

C:\DF\Digital_Forensics_Lab>cd Experiment-01-Ftk-Imager

C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager>dir
Volume in drive C is Windows
Volume Serial Number is 0A2E-2A00

Directory of C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager

27-08-2026  09:14 PM    <DIR>          .
27-08-2026  09:14 PM    <DIR>          ..
27-08-2026  10:06 PM    <DIR>          Results
               0 File(s)                0 bytes
               3 Dir(s)  90,353,725,440 bytes free

C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager>cd Results

C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager\Results>dir
Volume in drive C is Windows
Volume Serial Number is 0A2E-2A00

Directory of C:\DF\Digital_Forensics_Lab\Experiment-01-Ftk-Imager\Results

27-08-2026  10:06 PM    <DIR>          .
27-08-2026  09:14 PM    <DIR>          ..
27-08-2026  09:15 PM      20,971,520 Results.001
27-08-2026  09:37 PM      119,491 Results.001.txt
27-08-2026  09:15 PM      20,971,520 Results.002
27-08-2026  09:15 PM      20,971,520 Results.003
27-08-2026  09:15 PM      20,971,520 Results.004
27-08-2026  09:15 PM      20,971,520 Results.005
27-08-2026  09:15 PM      20,971,520 Results.006
27-08-2026  09:15 PM      20,971,520 Results.007
27-08-2026  09:15 PM      20,971,520 Results.008
27-08-2026  09:15 PM      20,971,520 Results.009
27-08-2026  09:15 PM      20,971,520 Results.010
27-08-2026  09:15 PM      20,971,520 Results.011
27-08-2026  09:15 PM      20,971,520 Results.012
27-08-2026  09:15 PM      20,971,520 Results.013
27-08-2026  09:15 PM      20,971,520 Results.014
27-08-2026  09:15 PM      20,971,520 Results.015
27-08-2026  09:15 PM      20,971,520 Results.016
27-08-2026  09:15 PM      20,971,520 Results.017

```

Part C: Volatile Memory Analysis using Volatility CLI / Workbench

Step 1: Launch Volatility Environment

Open PassMark Volatility Workbench (which acts as a GUI frontend for vol.exe).

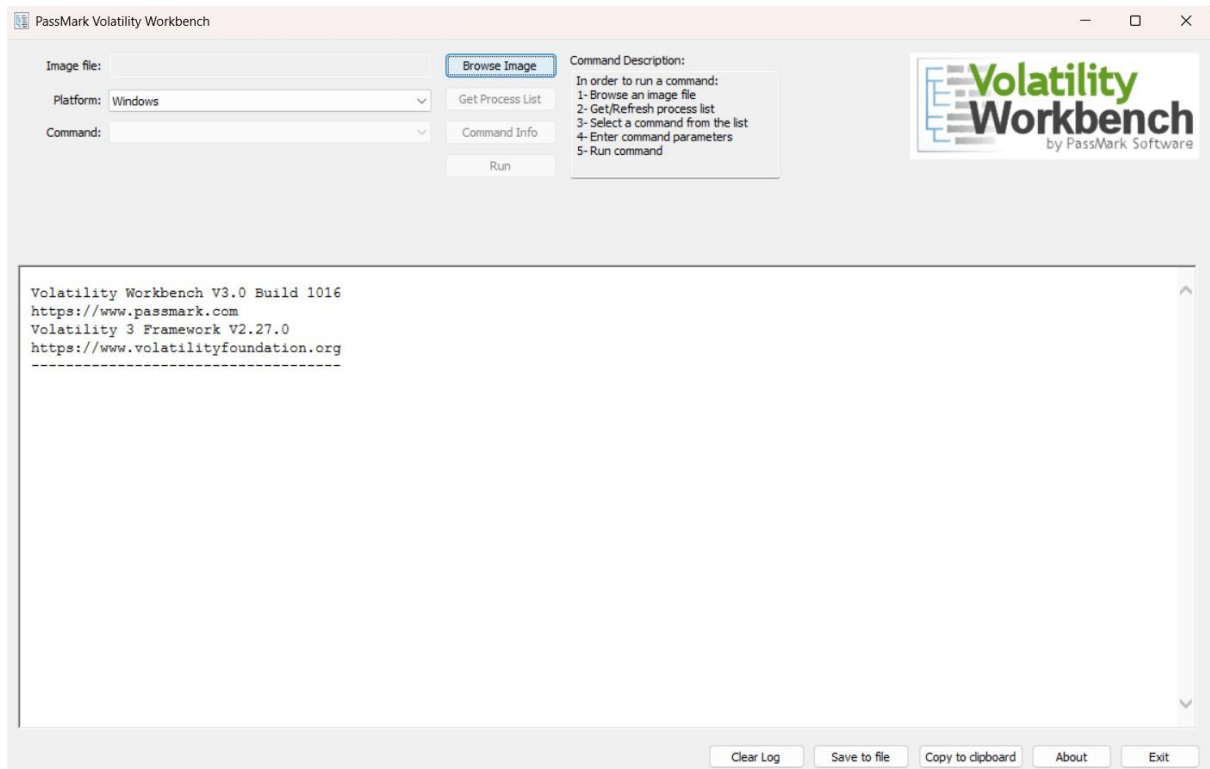


Figure 16: The default interface of PassMark Volatility Workbench

Step 2: Construct and Execute the Volatility Command

To parse the captured memory and list the active processes, load A:\DF\Inba.raw via **Browse Image**, select the **Windows** platform, choose windows.pslist.PsList from the dropdown, and click **Run**.

The console executes the following command in the background:

DOS

```
"C:\Users\tamil\Downloads\VolatilityWorkbench\vol.exe" -f "A:\DF\Inba.raw"
windows.pslist.PsList
```

Or

Use the workbench

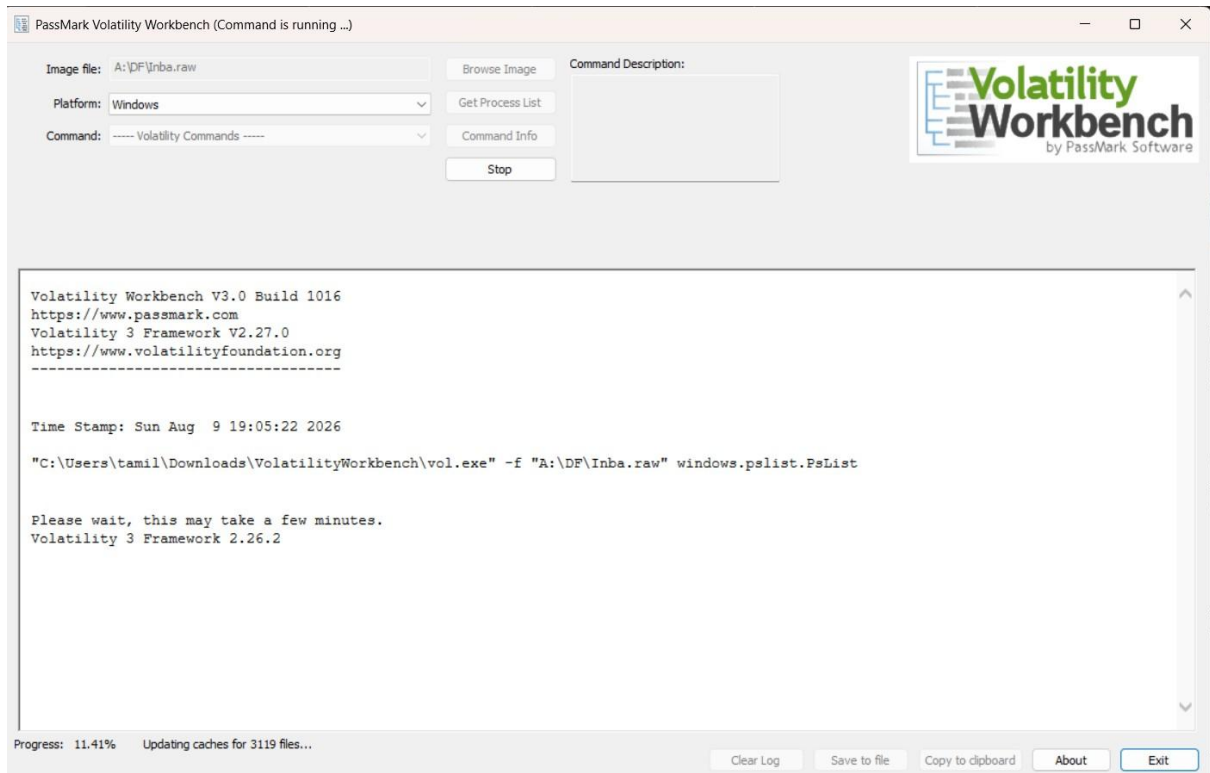


Figure 17: Execution of Volatility for the Inba.raw image

5. OBSERVATIONS & RESULTS

1. FTK Imager Cryptographic Verification

FTK Imager generated a summary report mathematically confirming the non-volatile image data integrity:

- **MD5 Hash:** 1f8957e555753455d076e144a74afef4 : verified
- **SHA1 Hash:** 613f621a072747853feb89ed81700c749c2cc7b1 : verified

2. Memory Capture & CLI Parsing

The Inba.raw file was successfully written using the winpmem_mini_x64_rc2.exe command. When analyzed, Volatility parsed the _EPROCESS blocks in kernel memory. The resulting output log accurately detailed the Process IDs (PID), Parent Process IDs (PPID), Image File Names, and timestamps of all active programs:

```

Volatility Workbench V3.0 Build 1016
https://www.passmark.com
Volatility 3 Framework V2.27.0
https://www.volatilityfoundation.org

Time Stamp: Sun Aug 9 19:05:22 2026

"C:\Users\tamil\Downloads\VolatilityWorkbench\vol.exe" -f "A:\DF\Inba.ram" windows.plist.PsList

Please wait, this may take a few minutes.
Volatility 3 Framework 2.26.2

```

PID	PPID	ImageFileName	Offset (V)	Threads	Handles	SessionId	Wow64	CreateTime	ExitTime	File output
4	0	System	0xbbb0d56ba040 813	-	N/A	False	2026-08-09 12:23:39.000000 UTC	N/A	Disabled	
188	4	Secure System	0xbbb0d5874040	0	-	N/A	False	2026-08-09 12:23:36.000000 UTC	N/A	Disabled
232	4	Registry	0xbbb0d5cae080	4	-	N/A	False	2026-08-09 12:23:36.000000 UTC	N/A	Disabled
740	4	csrss.exe	0xbbb0d5fac080	2	-	N/A	False	2026-08-09 12:23:39.000000 UTC	N/A	Disabled
924	964	csrss.exe	0xbbb0d5f1b080	13	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1060	964	wininit.exe	0xbbb0d5ef2100	3	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1068	1048	csrss.exe	0xbbb0d5f1b780	15	-	1	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1132	1048	winlogon.exe	0xbbb0d5f1c180	5	-	1	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1208	1060	services.exe	0xbbb0d5a28080	7	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1222	1060	lsass.exe	0xbbb0d5a280c0	1	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1240	1060	lsass.exe	0xbbb0d5a280c0	10	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1376	1208	svchost.exe	0xbbb0d5b5d080	17	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1388	1208	WUDFHost.exe	0xbbb0d5ba0c0	25	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1444	1060	fontdrvhost.exe	0xbbb0d5b73080	5	-	0	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1452	1132	fontdrvhost.exe	0xbbb0d5b74080	5	-	1	False	2026-08-09 12:23:46.000000 UTC	N/A	Disabled
1536	1208	svchost.exe	0xbbb0d5ab3080	10	-	0	False	2026-08-09 12:23:47.000000 UTC	N/A	Disabled
1620	1208	WUDFHost.exe	0xbbb0d5c99080	12	-	0	False	2026-08-09 12:23:47.000000 UTC	N/A	Disabled
1696	1208	WUDFHost.exe	0xbbb0d5c9b7080	5	-	0	False	2026-08-09 12:23:47.000000 UTC	N/A	Disabled
1804	1132	dmn.exe	0xbbb0d5e3e080 20	-	1	False	2026-08-09 12:23:47.000000 UTC	N/A	Disabled	
1872	1208	svchost.exe	0xbbb0d5e78080	57	-	0	False	2026-08-09 12:23:47.000000 UTC	N/A	Disabled

Figure 18: Volatility Workbench execution log displaying the extracted Process List

Note: The log accurately shows winpmem_mini_x (PID 19448) actively running, reflecting the acquisition tool executing at the time of the capture.

6. CONCLUSION

A complete digital forensics acquisition and analysis workflow was successfully executed via a combination of GUI and CLI tools. The target physical USB drive was securely imaged and verified using FTK Imager. Simultaneously, the system's live RAM was extracted via the Command Prompt using WinPmem. Finally, Volatility 3 command-line syntax was utilized to parse the raw memory file, successfully recovering critical live-state artifacts, including the process list and execution timestamps.